

Fast Authentication Between UAVs



Introduction:

- UAVs are widely used in surveillance, delivery and defence – all need secure communication.
- Authentication between UAVs and the Ground Station prevents impersonation, replay and cloning attacks.
- Physical Unclonable Functions (PUFs) give hardware-based identity **without storing any key** – ideal for resource-constrained UAVs.

Motivation:

- RSA/ECC need $5\text{--}10\times$ more computation & energy – unsuitable for battery-powered UAVs.
- Stored secret keys are vulnerable when a UAV is physically captured.
- Need a **fast peer-to-peer** authentication mechanism for UAV swarms, independent of the Ground Station.

Existing work / Literature review:

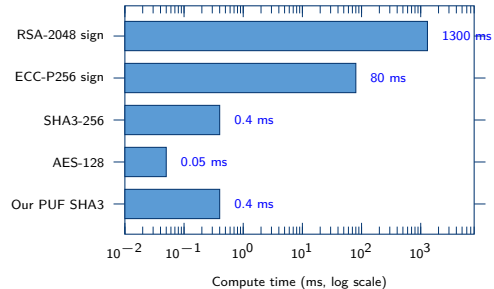


- PKI / symmetric protocols give strong security but **high compute overhead** – impractical for small UAVs.
- Lightweight schemes use hash-and-XOR (SPONGENT, SHA-3) to cut compute by **50–90%**.
- PUF-based methods derive identity from **manufacturing variations** – no key stored on the UAV.

Gaps in existing work:

- RSA / ECC need too much CPU for small UAVs.
- Stored-key protocols leak secrets when UAV is captured.
- Few schemes give **fast peer-to-peer** UAV auth without GS.

Compute time on Cortex-M4 @ 168 MHz (per auth)



RSA/ECC: wolfSSL Cortex-M4 benchmarks. SHA3/AES: Keccak Team

SUPERCOP & mbedTLS reports.

Work Done till date:



My work:

- Implemented 4-phase PUF auth in OMNeT++ 6.3 (10 UAVs + 1 GS); Arbiter PUF + BCH(255,131,18) fuzzy extractor; SHA3-256 / SPONGENT-160 modes.
- End-to-end benchmark: compute, network, energy, area; predicted Pi-4 / Cortex-M4 / ASIC numbers.

SHA3-256 vs SPONGENT-160 – detailed comparison:

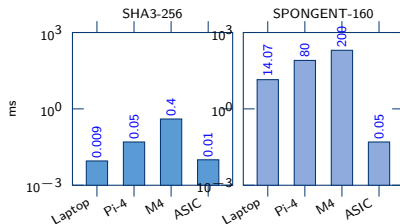
Metric	SHA3-256	SPONGENT-160
Gate area (130 nm) [1,2]	~15,000 GE	1,329 GE
Energy / hash (HW) [1,2]	~2.0 nJ	~0.66 nJ
Power (HW @100 kHz) [1]	~30 μ W	~1.6 μ W
Throughput (HW @100 kHz)	8 kbps	0.81 kbps
Compute – Laptop x86-64 (ours)	0.009 ms	14.07 ms
Compute – RaspPi 4 (A72) [3]	~0.05 ms	~80 ms
Compute – Cortex-M4 [3]	~0.4 ms	~200 ms
ASIC HW estimate [1,2]	~0.01 ms	~0.05 ms
Network delay (ours, OMNeT++)	0.665 / 0.467 ms	0.665 / 0.467 ms

[1] Bogdanov et al., SPONGENT, CHES'11. [2] FIPS-202. [3] wolfSSL bench.

Key summary:

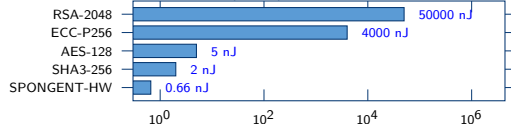
- SHA3-256 fastest in software; SPONGENT-160 11 $\times$ smaller silicon & 3 $\times$ less energy/hash (built for ASIC).
- Both use ~1000 $\times$ less energy than RSA-2048, ~80 $\times$ less than ECC-P256.
- Total auth = compute + network (OMNeT++ stadium, 10 UAVs):
 - P2 (UAV $\leftrightarrow$ GS): SHA3 0.007+0.665=0.67 ms; SPONGENT 6.70+0.665=7.37 ms.
 - P3 (UAV $\leftrightarrow$ UAV): SHA3 0.009+0.467=0.48 ms; SPONGENT 14.07+0.467=14.54 ms.
- 100% success, 430 B payload, no key storage; no MAC-layer contention modelled (lower bound).

Compute time per hash across devices (ms, log)



Values in ms. SPONGENT wins only in silicon (Bogdanov, CHES'11).

Energy per auth on UAV (nJ, log) – vs other crypto



RSA/ECC/AES: wolfSSL Cortex-M bench.